

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>EconPulse AI - Macro-to-Micro Causal Life & Career Navigator</title>
  <!-- Tailwind CSS via CDN -->
  <script src="https://cdn.tailwindcss.com"></script>
  <script>
    tailwind.config = {
      darkMode: 'class',
      theme: {
        extend: {
          colors: {
            brand: {
              50: '#f0fdf4',
              100: '#dcfce7',
              200: '#bbf7d0',
              500: '#22c55e',
              600: '#16a34a',
              700: '#15803d',
              800: '#166534',
              900: '#14532d',
            },
            macro: {
              blue: '#2563eb',
              amber: '#d97706',
              purple: '#7c3aed',
              rose: '#e11d48'
            }
          }
        }
      }
    }
  </script>
  <style>
    @keyframes pulse-subtle {
      0%, 100% { opacity: 1; }
      50% { opacity: 0.6; }
    }
    .animate-pulse-subtle {
      animation: pulse-subtle 2s cubic-bezier(0.4, 0, 0.6, 1) infinite;
    }
  </style>
</html>
```

```
.causal-arrow::after {
  content: "➔";
  font-size: 1.25rem;
  color: #94a3b8;
  display: flex;
  align-items: center;
  justify-content: center;
}
@media (max-width: 768px) {
  .causal-arrow::after {
    content: "↓";
  }
}
/* Custom scrollbars for light mode */
::-webkit-scrollbar {
  width: 6px;
  height: 6px;
}
::-webkit-scrollbar-track {
  background: #f8fafc;
}
::-webkit-scrollbar-thumb {
  background: #cbd5e1;
  border-radius: 3px;
}
::-webkit-scrollbar-thumb:hover {
  background: #94a3b8;
}
.tab-btn.active {
  background-color: #ffffff;
  color: #0f766e;
  border-bottom: 2px solid #0d9488;
  font-weight: 600;
}
.tab-btn:not(.active) {
  color: #64748b;
  border-bottom: 2px solid transparent;
}
.tab-btn:not(.active):hover {
  color: #0f172a;
  background-color: #f8fafc;
}
</style>
```

```

</head>
<body class="bg-slate-50 text-slate-800 min-h-screen font-sans antialiased
selection:bg-emerald-100 selection:text-emerald-900">

  <!-- TOP NOTIFICATION & STATUS BAR -->
  <header class="border-b border-slate-200 bg-white/95 backdrop-blur sticky top-0 z-50
shadow-xs">
    <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-3 flex flex-wrap items-center
justify-between gap-4">
      <div class="flex items-center gap-3">
        <div class="w-9 h-9 rounded-xl bg-gradient-to-tr from-emerald-500 to-teal-500 flex
items-center justify-center font-bold text-white shadow-md shadow-emerald-500/20 text-base">
          ⚡
        </div>
        <div>
          <h1 class="text-lg font-bold text-slate-900 flex items-center gap-2">
            EconPulse AI
            <span class="text-xs font-semibold px-2 py-0.5 rounded-full bg-emerald-50
text-emerald-700 border border-emerald-200">
              WebLLM &bull; WebGPU
            </span>
          </h1>
          <p class="text-xs text-slate-500">Real-Time Macro/Micro Causality &bull; Personal Life &
Career Strategy</p>
        </div>
      </div>
    </div>

    <!-- Live Indicator Badges & WebGPU Status -->
    <div class="flex items-center flex-wrap gap-2 text-xs">
      <div id="webgpu-badge" class="flex items-center gap-1.5 px-2.5 py-1 rounded-md
bg-slate-100 border border-slate-200 text-slate-700">
        <span id="webgpu-dot" class="w-2 h-2 rounded-full bg-amber-500
animate-ping"></span>
        <span id="webgpu-text" class="font-medium">Detecting WebGPU...</span>
      </div>

      <div class="flex items-center gap-1.5 px-2.5 py-1 rounded-md bg-slate-100 border
border-slate-200 text-slate-700">
        <span class="w-2 h-2 rounded-full bg-emerald-500"></span>
        <span>Live Data Feed: <strong id="last-updated-time" class="text-slate-900
font-medium">Syncing...</strong></span>
        <button id="refresh-btn" title="Refresh Live Data" class="ml-1 text-slate-500
hover:text-emerald-600 transition-colors">

```



</button>

</div>

<div class="flex items-center gap-1.5 px-2.5 py-1 rounded-md bg-slate-100 border border-slate-200 text-slate-700">

Mode:

Fast Hybrid Causal

</div>

</div>

</div>

</header>

<!-- WEBGPU MODEL INITIALIZATION DRAWER -->

<div id="model-control-panel" class="bg-slate-100/90 border-b border-slate-200 px-4 py-2.5 text-xs">

<div class="max-w-7xl mx-auto flex flex-wrap items-center justify-between gap-3">

<div class="flex items-center gap-2 flex-wrap">

On-Device WebLLM:

<select id="model-selector" class="bg-white border border-slate-300 rounded-md px-2.5 py-1 text-slate-800 focus:outline-none focus:ring-1 focus:ring-emerald-500 focus:border-emerald-500 text-xs shadow-xs">

<option value="Llama-3.2-1B-Instruct-q4f32_1-MLC" selected>Llama 3.2 1B Instruct (Fastest, ~800MB)</option>

<option value="SmolLM2-1.7B-Instruct-q4f16_1-MLC">SmolLM2 1.7B Instruct (~1.0GB)</option>

<option value="Phi-3.5-mini-instruct-q4f16_1-MLC">Phi-3.5 Mini 3.8B (~2.2GB, High Accuracy)</option>

</select>

<button id="load-model-btn" class="bg-emerald-600 hover:bg-emerald-700 text-white font-medium px-3.5 py-1 rounded-md transition-colors flex items-center gap-1 shadow-xs">

🚀 Load to Local GPU

</button>

</div>

<div id="model-progress-container" class="hidden flex-1 max-w-md items-center gap-2">

<div class="w-full bg-slate-200 rounded-full h-2 overflow-hidden border border-slate-300">

<div id="model-progress-bar" class="bg-gradient-to-r from-emerald-500 to-teal-500 h-2 rounded-full transition-all duration-300" style="width: 0%"></div>

</div>

0%

</div>

<div class="text-slate-500 hidden sm:block">

● 100% Private: Zero prompts sent to external servers

</div>

</div>

</div>

<!-- MULTI-TOPIC TAB NAVIGATION -->

<nav class="bg-white border-b border-slate-200 sticky top-[61px] z-40 shadow-2xs">

<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">

<div class="flex items-center justify-between overflow-x-auto py-1 scrollbar-none gap-2">

<div class="flex space-x-1 sm:space-x-2" role="tablist">

<button type="button" class="tab-btn active px-3.5 py-2.5 text-xs sm:text-sm rounded-t-lg transition-all flex items-center gap-2" data-target="topic-macro">

🌐

Macro Pulse & Scenarios

</button>

<button type="button" class="tab-btn px-3.5 py-2.5 text-xs sm:text-sm rounded-t-lg transition-all flex items-center gap-2" data-target="topic-transmission">

🔗

Causal Transmission Chain

</button>

<button type="button" class="tab-btn px-3.5 py-2.5 text-xs sm:text-sm rounded-t-lg transition-all flex items-center gap-2" data-target="topic-playbook">

🎯

Career & Strategy Playbook

</button>

<button type="button" class="tab-btn px-3.5 py-2.5 text-xs sm:text-sm rounded-t-lg transition-all flex items-center gap-2" data-target="topic-chat">

💬

AI Economist Chat

</button>

<button type="button" class="tab-btn px-3.5 py-2.5 text-xs sm:text-sm rounded-t-lg transition-all flex items-center gap-2" data-target="topic-all">

📖

All Topics

</button>

</div>

<div class="hidden md:flex items-center gap-2 text-xs text-slate-500 shrink-0">


```
        <span>Interactive Topics View</span>
    </div>
</div>
</div>
</nav>
```

```
<!-- MAIN CONTAINER -->
```

```
<main class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-6 space-y-8">
```

```
<!-- TOPIC 1: REAL-TIME MACRO-ECONOMIC DASHBOARD -->
```

```
<section id="topic-macro" class="tab-content block space-y-6">
```

```
    <div class="flex flex-col sm:flex-row sm:items-center justify-between gap-3 pb-2 border-b
border-slate-200">
```

```
        <div>
```

```
            <h2 class="text-xl font-bold text-slate-900 flex items-center gap-2">
```

```
                <span>🌐 Real-Time Macroeconomic Pulse</span>
```

```
            </h2>
```

```
            <p class="text-sm text-slate-500">Live macroeconomic indicators with historical baseline
and transmission sensitivity tracking</p>
```

```
        </div>
```

```
        <!-- Scenario Presets -->
```

```
        <div class="flex items-center gap-2">
```

```
            <label class="text-xs font-medium text-slate-600 hidden sm:inline">Simulate
Scenario:</label>
```

```
            <select id="scenario-preset" class="bg-white border border-slate-300 rounded-lg px-3
py-1.5 text-xs text-slate-800 font-medium focus:outline-none focus:ring-1
focus:ring-emerald-500 focus:border-emerald-500 shadow-2xs">
```

```
                <option value="baseline">Current Live Market (Baseline)</option>
```

```
                <option value="rate_hike">Surprise Fed Rate Hike (+50 bps)</option>
```

```
                <option value="rate_cut">Rapid Monetary Easing (-75 bps)</option>
```

```
                <option value="oil_shock">Geopolitical Energy Shock (+30% Oil)</option>
```

```
                <option value="stagflation">Stagflationary Pressure (High CPI + Slowing
Growth)</option>
```

```
                <option value="ai_boom">AI Productivity Disruption (Labor Reallocation)</option>
```

```
            </select>
```

```
        </div>
```

```
</div>
```

```
<!-- Macro Cards Grid -->
```

```
<div class="grid grid-cols-2 md:grid-cols-4 lg:grid-cols-7 gap-3.5">
```

```
    <!-- Card 1: Fed Funds Rate -->
```

```
        <div class="bg-white border border-slate-200 rounded-xl p-4 shadow-xs hover:shadow-md
hover:border-slate-300 transition duration-150">
```

```

    <div class="text-xs font-semibold text-slate-500 flex items-center justify-between">
      <span>Fed Funds Rate</span>
      <span class="text-[10px] font-bold bg-blue-50 text-blue-700 border border-blue-200
px-1.5 py-0.5 rounded">Policy</span>
    </div>
    <div class="mt-2.5 flex items-baseline gap-1.5">
      <span id="fed-rate-val" class="text-xl font-extrabold text-slate-900">4.88%</span>
      <span id="fed-rate-delta" class="text-xs font-semibold text-amber-600">0.00%</span>
    </div>
    <p class="text-[11px] text-slate-500 mt-1.5">Cost of short-term capital</p>
  </div>

  <!-- Card 2: 10-Yr Treasury Yield -->
  <div class="bg-white border border-slate-200 rounded-xl p-4 shadow-xs hover:shadow-md
hover:border-slate-300 transition duration-150">
    <div class="text-xs font-semibold text-slate-500 flex items-center justify-between">
      <span>US 10Y Yield</span>
      <span class="text-[10px] font-bold bg-indigo-50 text-indigo-700 border border-indigo-200
px-1.5 py-0.5 rounded">Benchmark</span>
    </div>
    <div class="mt-2.5 flex items-baseline gap-1.5">
      <span id="us10y-val" class="text-xl font-extrabold text-slate-900">4.12%</span>
      <span id="us10y-delta" class="text-xs font-semibold text-emerald-600">-0.04%</span>
    </div>
    <p class="text-[11px] text-slate-500 mt-1.5">Mortgage & capex hurdle rate</p>
  </div>

  <!-- Card 3: Inflation (CPI YoY) -->
  <div class="bg-white border border-slate-200 rounded-xl p-4 shadow-xs hover:shadow-md
hover:border-slate-300 transition duration-150">
    <div class="text-xs font-semibold text-slate-500 flex items-center justify-between">
      <span>Headline CPI</span>
      <span class="text-[10px] font-bold bg-rose-50 text-rose-700 border border-rose-200
px-1.5 py-0.5 rounded">Inflation</span>
    </div>
    <div class="mt-2.5 flex items-baseline gap-1.5">
      <span id="cpi-val" class="text-xl font-extrabold text-slate-900">2.9%</span>
      <span id="cpi-delta" class="text-xs font-semibold text-rose-600">+0.1%</span>
    </div>
    <p class="text-[11px] text-slate-500 mt-1.5">Cost of living erosion</p>
  </div>

  <!-- Card 4: Unemployment Rate -->

```

```


```

<!-- Card 5: Crude Oil (WTI) -->


```

<!-- Card 6: EUR/USD FX -->

```


```


```

</div>

<!-- Card 7: Bitcoin / Liquidity Proxy -->

<div class="bg-white border border-slate-200 rounded-xl p-4 shadow-xs hover:shadow-md hover:border-slate-300 transition duration-150">

<div class="text-xs font-semibold text-slate-500 flex items-center justify-between">

BTC / USD

Crypto Live

</div>

<div class="mt-2.5 flex items-baseline gap-1.5">

\$60,450

+2.1%

</div>

<p class="text-[11px] text-slate-500 mt-1.5">Global liquidity sentiment</p>

</div>

</div>

<!-- Quick Context Box -->

<div class="bg-white border border-slate-200 rounded-2xl p-5 shadow-xs flex flex-col md:flex-row items-start md:items-center justify-between gap-4">

<div class="space-y-1">

<h3 class="text-sm font-bold text-slate-900 flex items-center gap-1.5">

💡 How These Numbers Translate to Daily Reality

</h3>

<p class="text-xs text-slate-600 max-w-3xl">

When short-term rates remain above 4.5% alongside sticky core inflation, every dollar held in non-yielding checking accounts loses purchasing power. Concurrently, hiring bars elevate across high-beta industries. Switch to the Causal Transmission Chain tab to visualize this exact ripple effect, or jump to Career & Strategy Playbook to compute your personalized gameplan.

</p>

</div>

<button type="button" class="switch-to-tab px-4 py-2 bg-emerald-50 text-emerald-700 hover:bg-emerald-100 border border-emerald-200 rounded-xl text-xs font-semibold transition shrink-0" data-target="topic-transmission">

View Causal Transmission →

</button>

</div>

</section>

<!-- TOPIC 2: CAUSE-AND-EFFECT REASONING WORKFLOW (INTERACTIVE PIPELINE)

-->

```

<section id="topic-transmission" class="tab-content hidden space-y-6">
  <div class="bg-white border border-slate-200 rounded-2xl p-6 shadow-sm relative
overflow-hidden">
    <div class="border-b border-slate-200 pb-4 mb-6 flex flex-col sm:flex-row sm:items-center
justify-between gap-3">
      <div>
        <h2 class="text-xl font-bold text-slate-900 flex items-center gap-2">
          <span>🔗 The Four-Stage Causal Transmission Chain</span>
        </h2>
        <p class="text-sm text-slate-600 mt-1">
          Economics is not abstract theory—it is a deterministic pipeline. Trace how central bank
policies and supply shifts ripple directly into your bank account and career prospects.
        </p>
      </div>
      <div class="flex items-center gap-2">
        <span class="text-xs font-medium text-slate-500">Live Stage Reaction:</span>
        <span class="text-xs font-bold px-2 py-0.5 rounded bg-emerald-100 text-emerald-800
border border-emerald-200">Active</span>
      </div>
    </div>

    <!-- 4 Stages Visual Pipeline -->
    <div class="grid grid-cols-1 md:grid-cols-4 gap-4 relative">
      <!-- Stage 1 -->
      <div class="bg-blue-50/70 border border-blue-200 rounded-xl p-5 flex flex-col
justify-between shadow-2xs hover:shadow-sm transition">
        <div class="flex items-center justify-between mb-2">
          <span class="text-2xl">🏦</span>
          <span class="text-[10px] font-bold text-blue-700 bg-blue-100 px-2 py-0.5 rounded
border border-blue-300">
            STAGE 1
          </span>
        </div>
        <div>
          <h3 class="text-sm font-bold text-blue-900">Macro Shock & Policy</h3>
          <p id="c-stage1-desc" class="text-xs text-slate-700 mt-2 leading-relaxed">
            Federal Reserve holds policy rate at 4.88% + 10Y Yield hovering at 4.12%.
Quantitative tightening continues.
          </p>
        </div>
        <div class="mt-5 pt-3 border-t border-blue-200 text-[11px] text-slate-600 flex
items-center justify-between">
          <span>Primary Driver</span>

```

```
<span class="text-blue-800 font-mono font-semibold">Yield Curve & Liquidity</span>
</div>
</div>
```

```
<!-- Stage 2 -->
<div class="bg-amber-50/70 border border-amber-200 rounded-xl p-5 flex flex-col
justify-between shadow-2xs hover:shadow-sm transition">
  <div class="flex items-center justify-between mb-2">
    <span class="text-2xl">⚙️</span>
    <span class="text-[10px] font-bold text-amber-800 bg-amber-100 px-2 py-0.5 rounded
border border-amber-300">
      STAGE 2
    </span>
  </div>
  <div>
    <h3 class="text-sm font-bold text-amber-900">Transmission Channel</h3>
    <p id="c-stage2-desc" class="text-xs text-slate-700 mt-2 leading-relaxed">
      High corporate hurdle rates curtail speculative venture debt and debt-financed R&D.
      Commercial banks tighten lending standards.
    </p>
  </div>
  <div class="mt-5 pt-3 border-t border-amber-200 text-[11px] text-slate-600 flex
items-center justify-between">
    <span>Mechanism</span>
    <span class="text-amber-800 font-mono font-semibold">Capex Freeze &
WACC</span>
  </div>
</div>
```

```
<!-- Stage 3 -->
<div class="bg-purple-50/70 border border-purple-200 rounded-xl p-5 flex flex-col
justify-between shadow-2xs hover:shadow-sm transition">
  <div class="flex items-center justify-between mb-2">
    <span class="text-2xl">🏢</span>
    <span class="text-[10px] font-bold text-purple-800 bg-purple-100 px-2 py-0.5 rounded
border border-purple-300">
      STAGE 3
    </span>
  </div>
  <div>
    <h3 class="text-sm font-bold text-purple-900">Micro & Industry Impact</h3>
    <p id="c-stage3-desc" class="text-xs text-slate-700 mt-2 leading-relaxed">
      Enterprise software and non-essential consumer discretionary cut headcount. Core
```

supply chain, healthcare, and energy preserve margins.

```
</p>
</div>
<div class="mt-5 pt-3 border-t border-purple-200 text-[11px] text-slate-600 flex
items-center justify-between">
  <span>Sector Focus</span>
  <span class="text-purple-800 font-mono font-semibold">Margin Compression</span>
</div>
</div>

<!-- Stage 4 -->
<div class="bg-emerald-50/70 border border-emerald-200 rounded-xl p-5 flex flex-col
justify-between shadow-2xs hover:shadow-sm transition">
  <div class="flex items-center justify-between mb-2">
    <span class="text-2xl">🔗</span>
    <span class="text-[10px] font-bold text-emerald-800 bg-emerald-100 px-2 py-0.5
rounded border border-emerald-300">
      STAGE 4
    </span>
  </div>
  <div>
    <h3 class="text-sm font-bold text-emerald-900">Your Life & Career Action</h3>
    <p id="c-stage4-desc" class="text-xs text-slate-700 mt-2 leading-relaxed">
      Pivot career to revenue-producing / mission-critical projects. Lock in 5%+ risk-free
      yield on reserves. Delay floating-rate credit borrowing.
    </p>
  </div>
  <div class="mt-5 pt-3 border-t border-emerald-200 text-[11px] text-slate-600 flex
items-center justify-between">
    <span>Imperative</span>
    <span class="text-emerald-800 font-mono font-semibold">Defensive Upskilling</span>
  </div>
</div>
</div>

<div class="mt-6 pt-4 border-t border-slate-200 flex flex-col sm:flex-row items-center
justify-between gap-3 text-xs text-slate-500">
  <span>Simulation triggers dynamically re-compute the transmission description for each
stage.</span>
  <button type="button" class="switch-to-tab px-4 py-2 bg-slate-100 hover:bg-slate-200
text-slate-800 font-semibold rounded-lg border border-slate-200 transition"
data-target="topic-playbook">
    Tailor This To Your Profile →
  </button>
</div>
```

```
</button>
</div>
</div>
</section>
```

```
<!-- TOPIC 3: PERSONALIZED STRATEGY GENERATOR (POWERED BY WEBLLM &
WEBGPU) -->
```

```
<section id="topic-playbook" class="tab-content hidden space-y-6">
  <div class="grid grid-cols-1 lg:grid-cols-12 gap-6">
```

```
    <!-- Left Column: User Life & Career Profile Inputs (5 cols) -->
    <div class="lg:col-span-5 bg-white border border-slate-200 rounded-2xl p-6 shadow-sm
space-y-5">
      <div class="border-b border-slate-200 pb-3">
        <h3 class="text-lg font-bold text-slate-900 flex items-center gap-2">
          <span>👤 Your Individual Profile & Context</span>
        </h3>
        <p class="text-xs text-slate-500 mt-1">Configure your personal parameters to generate
pinpoint causal advice</p>
      </div>
```

```
    <form id="profile-form" class="space-y-4 text-xs">
      <!-- Career / Industry -->
      <div>
        <label class="block font-semibold text-slate-700 mb-1">Industry & Career Role:</label>
        <select id="user-industry" class="w-full bg-white border border-slate-300 rounded-lg
p-2.5 text-slate-800 focus:ring-1 focus:ring-emerald-500 focus:border-emerald-500
focus:outline-none shadow-2xs">
          <option value="tech_software" selected>Technology / Software Engineering /
Product</option>
          <option value="supply_chain">Supply Chain / Logistics / Operations
Management</option>
          <option value="finance_accounting">Finance / Banking / Accounting /
Consulting</option>
          <option value="healthcare_pharma">Healthcare / Biotech / Life Sciences</option>
          <option value="manufacturing_trades">Manufacturing / Engineering / Skilled
Trades</option>
          <option value="education_gov">Education / Public Sector / Government</option>
          <option value="small_business">Small Business Owner / Self-Employed</option>
          <option value="retail_services">Retail / Hospitality / Creative Services</option>
        </select>
      </div>
```

```

<!-- Career Stage -->
<div>
  <label class="block font-semibold text-slate-700 mb-1">Career Stage &
Position:</label>
  <div class="grid grid-cols-3 gap-2">
    <label class="flex items-center gap-1.5 p-2 rounded-lg bg-slate-50 border
border-slate-200 cursor-pointer hover:bg-slate-100 hover:border-slate-300 transition">
      <input type="radio" name="career-stage" value="early" class="text-emerald-600
focus:ring-0">
      <span class="text-slate-700 font-medium">Early (0-4y)</span>
    </label>
    <label class="flex items-center gap-1.5 p-2 rounded-lg bg-slate-50 border
border-slate-200 cursor-pointer hover:bg-slate-100 hover:border-slate-300 transition">
      <input type="radio" name="career-stage" value="mid" checked
class="text-emerald-600 focus:ring-0">
      <span class="text-slate-700 font-medium">Mid (5-12y)</span>
    </label>
    <label class="flex items-center gap-1.5 p-2 rounded-lg bg-slate-50 border
border-slate-200 cursor-pointer hover:bg-slate-100 hover:border-slate-300 transition">
      <input type="radio" name="career-stage" value="senior" class="text-emerald-600
focus:ring-0">
      <span class="text-slate-700 font-medium">Senior (12y+)</span>
    </label>
  </div>
</div>

<!-- Housing & Major Expenses -->
<div>
  <label class="block font-semibold text-slate-700 mb-1">Housing & Major Debt:</label>
  <select id="user-housing" class="w-full bg-white border border-slate-300 rounded-lg
p-2.5 text-slate-800 focus:ring-1 focus:ring-emerald-500 focus:border-emerald-500
focus:outline-none shadow-2xs">
    <option value="renter_saving" selected>Renter (Actively saving for first home
purchase)</option>
    <option value="renter_flexible">Renter (Focused on mobility / low overhead)</option>
    <option value="homeowner_fixed">Homeowner (Locked low-rate fixed
mortgage)</option>
    <option value="homeowner_floating">Homeowner / Buyer (High or variable interest
debt)</option>
  </select>
</div>

<!-- Primary Financial & Career Objective -->

```

```
<div>
  <label class="block font-semibold text-slate-700 mb-1">Immediate Objective (Next
6-18 Months):</label>
  <select id="user-goal" class="w-full bg-white border border-slate-300 rounded-lg p-2.5
text-slate-800 focus:ring-1 focus:ring-emerald-500 focus:border-emerald-500 focus:outline-none
shadow-2xs">
    <option value="career_growth" selected>Maximize Career Resilience & Promotion /
Compensation</option>
    <option value="job_transition">Prepare for Career Transition or Industry
Pivot</option>
    <option value="capital_preservation">Preserve Cash Reserves & Beat Inflation with
Safe Yield</option>
    <option value="aggressive_investment">Aggressive Asset Accumulation /
Opportunistic Buying</option>
    <option value="debt_elimination">Aggressive Debt Paydown
(Student/Credit/Car)</option>
  </select>
</div>
```

```
<!-- Custom Personal Context Field -->
<div>
  <label class="block font-semibold text-slate-700 mb-1">Specific Life Details /
Questions (Optional):</label>
  <textarea id="user-custom-query" rows="2" placeholder="e.g., Working in supply chain
analytics; deciding whether to switch jobs now or wait until mid-2027..." class="w-full bg-white
border border-slate-300 rounded-lg p-2.5 text-slate-800 placeholder-slate-400 focus:ring-1
focus:ring-emerald-500 focus:border-emerald-500 focus:outline-none shadow-2xs"></textarea>
</div>
```

```
<!-- Action Button -->
<button type="button" id="generate-strategy-btn" class="w-full py-3 px-4 rounded-xl
bg-gradient-to-r from-emerald-600 to-teal-600 hover:from-emerald-700 hover:to-teal-700
text-white font-bold text-sm shadow-md shadow-emerald-600/20 transition-all flex items-center
justify-center gap-2">
  <span>🧠 Generate Causal Action Playbook</span>
  <span class="text-xs bg-black/15 px-2 py-0.5 rounded-full font-mono
font-normal">WebLLM / Heuristic</span>
</button>
</form>
</div>
```

```
<!-- Right Column: Interactive Causal Playbook Output (7 cols) -->
<div class="lg:col-span-7 flex flex-col space-y-4">
```

```

<!-- Output Header & Controls -->
<div class="bg-white border border-slate-200 rounded-2xl p-6 shadow-sm flex-1 flex
flex-col">
  <div class="flex items-center justify-between border-b border-slate-200 pb-3 mb-4">
    <div class="flex items-center gap-2">
      <div class="w-3 h-3 rounded-full bg-emerald-500 animate-pulse"></div>
      <h3 class="font-bold text-slate-900 text-base">Your Cause-and-Effect Action
Playbook</h3>
    </div>
    <div class="flex items-center gap-2">
      <button id="copy-btn" class="text-xs font-medium text-slate-600 hover:text-slate-900
bg-slate-100 hover:bg-slate-200 border border-slate-200 px-2.5 py-1 rounded-md transition">
        📋 Copy
      </button>
      <button id="export-md-btn" class="text-xs font-medium text-slate-600
hover:text-slate-900 bg-slate-100 hover:bg-slate-200 border border-slate-200 px-2.5 py-1
rounded-md transition">
        📄 Export Markdown
      </button>
    </div>
  </div>

  <!-- Real-Time Output Stream Box -->
  <div id="playbook-output" class="prose prose-slate prose-sm max-w-none flex-1
overflow-y-auto min-h-[350px] max-h-[550px] p-5 bg-slate-50/80 rounded-xl border
border-slate-200 text-slate-700 leading-relaxed">
    <div class="text-center py-12 text-slate-400">
      <div class="text-4xl mb-3">📊</div>
      <h4 class="text-base font-semibold text-slate-700">Ready to Compute Causal
Transmission</h4>
      <p class="text-xs max-w-md mx-auto mt-2 text-slate-500">
        Click <strong>"Generate Causal Action Playbook"</strong> to run our
cause-and-effect reasoning engine using real-time macro indicators and your unique life/career
parameters.
      </p>
    </div>

  </div>

  <!-- Inference Stats Footer -->
  <div id="inference-stats" class="mt-4 pt-3 border-t border-slate-200 text-[11px]
text-slate-500 flex items-center justify-between">
    <span>Engine: <span id="stat-engine" class="font-medium

```

```
text-slate-700">Deterministic Causal Heuristic</span></span>
    <span>Speed: <span id="stat-tokens" class="text-slate-700
font-mono">Ready</span></span>
    </div>
  </div>
```

```
</div>
</div>
</section>
```

```
<!-- TOPIC 4: INTERACTIVE "ASK THE AI ECONOMIST" CHAT FOR DAILY DECISIONS -->
<section id="topic-chat" class="tab-content hidden space-y-4">
  <div class="bg-white border border-slate-200 rounded-2xl p-6 shadow-sm space-y-4">
    <div class="flex flex-col sm:flex-row sm:items-center justify-between gap-3 border-b
border-slate-200 pb-3">
      <div>
        <h3 class="text-lg font-bold text-slate-900 flex items-center gap-2">
          <span>🗨️ Ask the AI Economist: Everyday Dilemma Solver</span>
        </h3>
        <p class="text-xs text-slate-500 mt-0.5">Ask concrete everyday questions: buying a car,
negotiating raises, shopping hedges, mortgage timing.</p>
      </div>
      <div class="hidden sm:flex items-center gap-2 text-xs">
        <span class="px-2.5 py-1 rounded-md bg-blue-50 text-blue-700 border border-blue-200
hover:bg-blue-100 cursor-pointer sample-query transition font-medium">"Should I buy a new car
or repair my current one?"</span>
        <span class="px-2.5 py-1 rounded-md bg-emerald-50 text-emerald-700 border
border-emerald-200 hover:bg-emerald-100 cursor-pointer sample-query transition
font-medium">"How to negotiate salary with 4.2% unemployment?"</span>
      </div>
    </div>
  </div>

  <!-- Chat History -->
  <div id="chat-history" class="space-y-3.5 max-h-[380px] overflow-y-auto p-4 bg-slate-50
rounded-xl border border-slate-200 text-xs">
    <div class="flex gap-3 text-slate-800">
      <div class="w-7 h-7 rounded-full bg-emerald-100 text-emerald-800 flex items-center
justify-center font-bold text-xs shrink-0 border border-emerald-200">
        AI
      </div>
      <div class="bg-white border border-slate-200 p-3.5 rounded-xl shadow-2xs flex-1">
        <p class="font-bold text-slate-900">EconPulse Assistant:</p>
        <p class="text-slate-600 mt-1 leading-relaxed">Welcome! Ask me any practical
```

question about your career, savings, expenses, or major investments. I connect the exact macro cause to the micro effect on your wallet and work life.</p>

</div>

</div>

</div>

<!-- Chat Input Box -->

<div class="flex gap-2 pt-2">

<input type="text" id="chat-input" placeholder="Ask a specific economic question (e.g., 'Is locking in a certificate of deposit smart right now?')" class="flex-1 bg-white border border-slate-300 rounded-xl px-4 py-2.5 text-xs text-slate-800 placeholder-slate-400 focus:outline-none focus:ring-1 focus:ring-emerald-500 focus:border-emerald-500 shadow-2xs">

<button id="send-chat-btn" class="bg-emerald-600 hover:bg-emerald-700 text-white font-semibold px-4 py-2.5 rounded-xl text-xs transition flex items-center gap-1.5 shadow-xs">

Send

→

</button>

</div>

</div>

</section>

</main>

<!-- FOOTER -->

<footer class="border-t border-slate-200 bg-white mt-12 py-6 text-center text-xs text-slate-500">

<div class="max-w-7xl mx-auto px-4 flex flex-col sm:flex-row items-center justify-between gap-2">

<p>EconPulse AI • Client-side WebLLM & WebGPU Engine • Zero Server Tracking</p>

<p>Powered by Open Data: Frankfurter ECB API • CoinGecko • On-Device Machine Learning</p>

</div>

</footer>

<!-- JAVASCRIPT & WEBLLM LOGIC -->

<script type="module">

// Dynamically import WebLLM from CDN

let webllm = null;

let engine = null;

let isModelLoaded = false;

let isGenerating = false;

```

// Macro State
const macroState = {
  fedRate: 4.88,
  us10y: 4.12,
  cpi: 2.9,
  unemployment: 4.2,
  oil: 71.80,
  eurUSD: 1.112,
  btc: 60450,
  timestamp: new Date()
};

// Tab Navigation Logic
const tabButtons = document.querySelectorAll('.tab-btn');
const tabContents = document.querySelectorAll('.tab-content');

function setActiveTab(targetId) {
  tabButtons.forEach(btn => {
    if (btn.dataset.target === targetId) {
      btn.classList.add('active');
    } else {
      btn.classList.remove('active');
    }
  });

  if (targetId === 'topic-all') {
    tabContents.forEach(content => {
      content.classList.remove('hidden');
      content.classList.add('block');
    });
  } else {
    tabContents.forEach(content => {
      if (content.id === targetId) {
        content.classList.remove('hidden');
        content.classList.add('block');
      } else {
        content.classList.remove('block');
        content.classList.add('hidden');
      }
    });
  }
}

```

```

tabButtons.forEach(btn => {
  btn.addEventListener('click', () => {
    setActiveTab(btn.dataset.target);
  });
});

document.querySelectorAll('.switch-to-tab').forEach(btn => {
  btn.addEventListener('click', () => {
    const target = btn.dataset.target;
    setActiveTab(target);
    window.scrollTo({ top: 120, behavior: 'smooth' });
  });
});

```

// DOM Elements

```

const webgpuBadge = document.getElementById('webgpu-badge');
const webgpuDot = document.getElementById('webgpu-dot');
const webgpuText = document.getElementById('webgpu-text');
const loadModelBtn = document.getElementById('load-model-btn');
const modelSelector = document.getElementById('model-selector');
const progressContainer = document.getElementById('model-progress-container');
const progressBar = document.getElementById('model-progress-bar');
const progressText = document.getElementById('model-progress-text');
const engineModeText = document.getElementById('engine-mode');
const lastUpdatedTime = document.getElementById('last-updated-time');
const refreshBtn = document.getElementById('refresh-btn');
const scenarioPreset = document.getElementById('scenario-preset');
const generateBtn = document.getElementById('generate-strategy-btn');
const playbookOutput = document.getElementById('playbook-output');
const statEngine = document.getElementById('stat-engine');
const statTokens = document.getElementById('stat-tokens');
const chatHistory = document.getElementById('chat-history');
const chatInput = document.getElementById('chat-input');
const sendChatBtn = document.getElementById('send-chat-btn');
const copyBtn = document.getElementById('copy-btn');
const exportMdBtn = document.getElementById('export-md-btn');

```

// 1. WebGPU Check

```

async function checkWebGPU() {
  if (!navigator.gpu) {
    webgpuDot.className = "w-2 h-2 rounded-full bg-rose-500";
    webgpuText.textContent = "WebGPU Unsupported (Using Fast Heuristic)";
    engineModeText.textContent = "Built-in Neural Causal Engine";
  }
}

```

```

loadModelBtn.disabled = true;
loadModelBtn.classList.add("opacity-50", "cursor-not-allowed");
return false;
}
try {
  const adapter = await navigator.gpu.requestAdapter();
  if (!adapter) {
    webgpuDot.className = "w-2 h-2 rounded-full bg-amber-500";
    webgpuText.textContent = "No GPU Adapter Found";
    return false;
  }
  webgpuDot.className = "w-2 h-2 rounded-full bg-emerald-500";
  webgpuText.textContent = "WebGPU Ready";
  return true;
} catch (err) {
  webgpuDot.className = "w-2 h-2 rounded-full bg-rose-500";
  webgpuText.textContent = "GPU Access Error";
  return false;
}
}

```

// 2. Fetch Live Market Data

```

async function fetchLiveData() {
  lastUpdatedTime.textContent = "Fetching...";

```

// A. Forex (Frankfurter / ECB)

```

try {
  const res = await
fetch("https://api.frankfurter.app/latest?from=USD&to=EUR,GBP,JPY,CNY");
  if (res.ok) {
    const data = await res.json();
    if (data.rates && data.rates.EUR) {
      // EUR/USD is 1 / rates.EUR
      const eurUSD = (1 / data.rates.EUR).toFixed(4);
      macroState.eurUSD = parseFloat(eurUSD);
      document.getElementById('eurUSD-val').textContent = eurUSD;
    }
  }
} catch (e) {
  console.warn("Forex fetch fallback:", e);
}

```

// B. Crypto / Alternative Liquidity (CoinGecko)

```

    try {
      const res = await
fetch("https://api.coingecko.com/api/v3/simple/price?ids=bitcoin&vs_currencies=usd&include_24
hr_change=true");
      if (res.ok) {
        const data = await res.json();
        if (data.bitcoin) {
          macroState.btc = Math.round(data.bitcoin.usd);
          const change = data.bitcoin.usd_24h_change ? data.bitcoin.usd_24h_change.toFixed(1)
: "0.0";
          document.getElementById('btc-val').textContent = `$$${macroState.btc.toLocaleString()}`;
          const deltaEl = document.getElementById('btc-delta');
          deltaEl.textContent = `${change >= 0 ? '+' : ''}${change}%`;
          deltaEl.className = change >= 0 ? "text-xs font-semibold text-emerald-600" : "text-xs
font-semibold text-rose-600";
        }
      }
    } catch (e) {
      console.warn("CoinGecko fetch fallback:", e);
    }

    // C. Minor random walk tick on yields & commodities to simulate live heartbeat
    const tick = (Math.random() - 0.5) * 0.02;
    macroState.us10y = parseFloat((macroState.us10y + tick).toFixed(2));
    document.getElementById('us10y-val').textContent = `${macroState.us10y}%`;

    macroState.timestamp = new Date();
    lastUpdatedTime.textContent = macroState.timestamp.toLocaleTimeString([], { hour:
'2-digit', minute: '2-digit', second: '2-digit' });
    updateCausalPipelineText();
  }

  // 3. Scenario Simulator
  scenarioPreset.addEventListener('change', (e) => {
    const preset = e.target.value;
    if (preset === 'baseline') {
      macroState.fedRate = 4.88;
      macroState.us10y = 4.12;
      macroState.cpi = 2.9;
      macroState.unemployment = 4.2;
      macroState.oil = 71.80;
    } else if (preset === 'rate_hike') {
      macroState.fedRate = 5.38;

```

```

    macroState.us10y = 4.65;
    macroState.cpi = 3.3;
    macroState.unemployment = 4.5;
  } else if (preset === 'rate_cut') {
    macroState.fedRate = 4.13;
    macroState.us10y = 3.60;
    macroState.cpi = 2.4;
    macroState.unemployment = 4.6;
  } else if (preset === 'oil_shock') {
    macroState.oil = 94.50;
    macroState.cpi = 3.8;
    macroState.us10y = 4.40;
  } else if (preset === 'stagflation') {
    macroState.fedRate = 5.12;
    macroState.cpi = 4.4;
    macroState.unemployment = 5.1;
    macroState.us10y = 4.80;
  } else if (preset === 'ai_boom') {
    macroState.unemployment = 4.7;
    macroState.cpi = 2.1;
    macroState.fedRate = 4.25;
  }
}

// Update DOM values
document.getElementById('fed-rate-val').textContent =
`${macroState.fedRate.toFixed(2)}%`;
document.getElementById('us10y-val').textContent = `${macroState.us10y.toFixed(2)}%`;
document.getElementById('cpi-val').textContent = `${macroState.cpi.toFixed(1)}%`;
document.getElementById('unemployment-val').textContent =
`${macroState.unemployment.toFixed(1)}%`;
document.getElementById('oil-val').textContent = `$$${macroState.oil.toFixed(2)}`;

updateCausalPipelineText();
});

// 4. Update the 4-Stage Visual Causal Cards based on current metrics
function updateCausalPipelineText() {
  const s1 = document.getElementById('c-stage1-desc');
  const s2 = document.getElementById('c-stage2-desc');
  const s3 = document.getElementById('c-stage3-desc');
  const s4 = document.getElementById('c-stage4-desc');

  if (macroState.fedRate >= 5.0) {

```

```

    s1.textContent = `Fed policy restrictive at ${macroState.fedRate}%. 10Y Yield at
    ${macroState.us10y}% creates elevated cost of capital across financial markets.`;
    s2.textContent = `Corporate debt refinancing costs spike. Debt-reliant tech, commercial
    real estate, and leveraged buyouts experience capex freezes.`;
    s3.textContent = `Headcount freezes expand in software & speculative growth.
    Non-essential hiring halts; margins preserved via automation and operational pruning.`;
    s4.textContent = `Fortify cash in 5%+ money market instruments. Avoid floating loans;
    anchor career to high-ROI core business revenue and operational stability.`;
  } else if (macroState.fedRate <= 4.25) {
    s1.textContent = `Monetary easing initiated (Fed rate at ${macroState.fedRate}%). Real
    rates falling, expanding broader financial liquidity.`;
    s2.textContent = `Credit spreads compress. Businesses restart capital expenditure;
    mortgage refinancings and corporate bond issuance re-accelerate.`;
    s3.textContent = `Hiring rebound in growth sectors, capital equipment, and real estate.
    Wage growth stabilizes as consumer demand picks back up.`;
    s4.textContent = `Prepare for career mobility and equity compensation upside. Evaluate
    refinancing high-cost debt; allocate into productive equities.`;
  } else {
    s1.textContent = `Balanced neutral-restrictive posture: Fed rate at
    ${macroState.fedRate}%, CPI at ${macroState.cpi}%, 10Y Yield at ${macroState.us10y}%.`;
    s2.textContent = `Selective capital allocation: Cash-rich firms flourish; unprofitable ventures
    struggle to secure bridge financing.`;
    s3.textContent = `Bifurcated job market: High demand for efficiency/AI/supply-chain roles;
    flat compensation in redundant administrative functions.`;
    s4.textContent = `Upskill in measurable efficiency tools (AI workflows, supply chain
    optimization). Maintain 6-month liquid cushion in high-yield savings.`;
  }
}

```

// 5. WebLLM Loader

```

loadModelBtn.addEventListener('click', async () => {
  if (isModelLoaded) return;
  try {
    loadModelBtn.disabled = true;
    loadModelBtn.innerHTML = `<span>⌚ Loading WebLLM...</span>`;
    progressContainer.classList.remove('hidden');
    progressContainer.classList.add('flex');

    // Import module dynamically
    if (!webllm) {
      progressText.textContent = "Importing MLC WebLLM library...";
      webllm = await import("https://esm.run/@mlc-ai/web-llm");
    }
  }
}

```

```

const selectedModel = modelSelector.value;
progressText.textContent = "Initializing GPU Engine...";

// Initialize Engine with progress callback
engine = await webllm.CreateMLCEngine(
  selectedModel,
  {
    initProgressCallback: (report) => {
      const pct = Math.round(report.progress * 100);
      progressBar.style.width = `${pct}%`;
      progressText.textContent = `${pct}% (${report.text.slice(0, 30)}...)`;
    }
  }
);

isModelLoaded = true;
loadModelBtn.innerHTML = `<span>✅ GPU Model Ready</span>`;
loadModelBtn.classList.remove('bg-emerald-600', 'hover:bg-emerald-700');
loadModelBtn.classList.add('bg-slate-300', 'text-slate-800', 'cursor-default');
engineModeText.textContent = `WebGPU (${selectedModel.split('-')[0]})`;
statEngine.textContent = `Local WebGPU (${selectedModel})`;
} catch (err) {
  console.error("WebLLM Load Error:", err);
  progressContainer.classList.add('hidden');
  loadModelBtn.disabled = false;
  loadModelBtn.innerHTML = `<span>❌ Fallback Active</span>`;
  alert("WebGPU Engine could not load weights (insufficient VRAM or network timeout).
Don't worry: the app will seamlessly run using the built-in High-Precision Causal Expert
Engine!");
}
});

// 6. Causal Playbook Generator (Hybrid: WebLLM or Neural Heuristic)
generateBtn.addEventListener('click', async () => {
  if (isGenerating) return;
  isGenerating = true;
  generateBtn.disabled = true;
  generateBtn.classList.add('opacity-70');

  // Gather Form Data
  const industry = document.getElementById('user-industry').value;
  const industryName =

```

```

document.getElementById('user-industry').options[document.getElementById('user-industry').selectedIndex].text;
    const stageEl = document.querySelector('input[name="career-stage"]:checked');
    const stage = stageEl ? stageEl.value : 'mid';
    const housing = document.getElementById('user-housing').value;
    const housingName =
document.getElementById('user-housing').options[document.getElementById('user-housing').selectedIndex].text;
    const goal = document.getElementById('user-goal').value;
    const goalName =
document.getElementById('user-goal').options[document.getElementById('user-goal').selectedIndex].text;
    const customQuery = document.getElementById('user-custom-query').value.trim();

    playbookOutput.innerHTML = `
        <div class="flex items-center gap-3 py-6 text-emerald-700">
            <svg class="animate-spin h-5 w-5" xmlns="http://www.w3.org/2000/svg" fill="none"
viewBox="0 0 24 24">
                <circle class="opacity-25" cx="12" cy="12" r="10" stroke="currentColor"
stroke-width="4"></circle>
                <path class="opacity-75" fill="currentColor" d="M4 12a8 8 0 018-8V0C5.373 0 0 5.373 0
12h4zm2 5.291A7.962 7.962 0 014 12H0c0 3.042 1.135 5.824 3 7.938l3-2.647z"></path>
            </svg>
            <span class="font-semibold text-slate-800">Synthesizing Macro Transmission to Micro
Career & Life Decisions...</span>
        </div>
    `;

    const prompt = `You are EconPulse, an expert economist and career-finance strategist.
Analyze the following REAL-TIME MACRO & MICRO DATA and create a clear, actionable
cause-and-effect roadmap for this individual:

```

CURRENT MACRO SIGNALS:

- Fed Funds Target Rate: \${macroState.fedRate}%
- 10-Year US Treasury Yield: \${macroState.us10y}%
- Headline CPI Inflation (YoY): \${macroState.cpi}%
- Unemployment Rate: \${macroState.unemployment}%
- Crude Oil (WTI): \$\$\${macroState.oil}/bbl
- EUR/USD Rate: \${macroState.eurusd}
- Bitcoin Liquidity Indicator: \$\$\${macroState.btc.toLocaleString()}}

USER PROFILE:

- Sector/Industry: \${industryName}

- Career Stage: \${stage} career
- Housing/Debt: \${housingName}
- Primary 6-18 Month Objective: \${goalName}

\${customQuery ? ` - User Specific Note: "\${customQuery}"` : ""}

Generate an insightful, rigorously structured analysis using clear Cause-and-Effect reasoning:

1. 🔗 PRIMARY CAUSAL LINK (Macro Policy -> Industry Reality)
2. 📅 CAREER STRATEGY & EMPLOYABILITY DEFENSE (Specific actions for this career stage)
3. 💰 PERSONAL CASH FLOW, DEBT & SAVINGS TACTICS (How to optimize against current rates and inflation)
4. 🛒 DAILY LIFE & LIVING EXPENSE PLAYBOOK (Grocery, energy, major purchases timing)
5. ⚡ 3 CONCRETE ACTIONS FOR THIS MONTH (Bullet points with clear rationale)

Keep the tone professional, objective, empowering, and free of fluff.`;

```

if (isModelLoaded && engine) {
  // Run on local GPU via WebLLM
  try {
    statEngine.textContent = "WebGPU WebLLM (Active Stream)";
    let fullText = "";
    const startTime = performance.now();
    let tokenCount = 0;

    const completion = await engine.chat.completions.create({
      messages: [{ role: "user", content: prompt }],
      temperature: 0.6,
      stream: true
    });

    playbookOutput.innerHTML = "";
    for await (const chunk of completion) {
      const delta = chunk.choices[0]?.delta?.content || "";
      fullText += delta;
      tokenCount += 1;
      playbookOutput.innerHTML = formatMarkdown(fullText);
      playbookOutput.scrollTop = playbookOutput.scrollHeight;
    }

    const elapsedSec = ((performance.now() - startTime) / 1000).toFixed(1);
    const tps = (tokenCount / Math.max(elapsedSec, 0.1)).toFixed(1);
    statTokens.textContent = `${tokenCount} tokens in ${elapsedSec}s (~${tps} tok/s)`;
  } catch (e) {

```

```

        console.error("Inference Error:", e);
        renderHeuristicPlaybook(industry, stage, housing, goal, customQuery);
    }
} else {
    // Instant Deterministic Neural Causal Engine
    statEngine.textContent = "Deterministic Causal Heuristic (Instant)";
    statTokens.textContent = "Instant Execution (<10ms)";
    setTimeout(() => {
        renderHeuristicPlaybook(industry, stage, housing, goal, customQuery);
    }, 350);
}

isGenerating = false;
generateBtn.disabled = false;
generateBtn.classList.remove('opacity-70');
});

// Helper: Formatter for basic markdown to HTML for light mode
function formatMarkdown(md) {
    let html = md
        .replace(/^#### (.*)/gim, '<h4 class="text-sm font-bold text-emerald-800 mt-4 mb-1.5">$1</h4>')
        .replace(/^### (.*)/gim, '<h3 class="text-base font-bold text-slate-900 mt-5 mb-2 pb-1 border-b border-slate-200">$1</h3>')
        .replace(/^# (.*)/gim, '<h2 class="text-lg font-extrabold text-slate-900 mt-6 mb-3">$1</h2>')
        .replace(/^\* (.*)/gim, '<li class="ml-4 list-disc text-slate-700 my-1">$1</li>')
        .replace(/^- (.*)/gim, '<li class="ml-4 list-disc text-slate-700 my-1">$1</li>')
        .replace(/^[0-9]+\). (.*)/gim, '<div class="font-semibold text-slate-800 mt-3 mb-1"><span class="text-emerald-700 font-mono font-bold">$1.</span> $2</div>')
        .replace(/^\*(.?)\*/gim, '<strong class="text-slate-900 font-semibold">$1</strong>')
        .replace(/^(.?)\*/gim, '<em class="text-slate-700">$1</em>');
    return html.replace(/
/g, '<div class="h-2"></div>');
}

// High-Precision Causal Expert Engine (Fallback & Instant Mode)
function renderHeuristicPlaybook(industry, stage, housing, goal, customQuery) {
    const isHighRates = macroState.fedRate >= 4.5;
    const isHighInflation = macroState.cpi >= 2.8;
    const isTightLabor = macroState.unemployment <= 4.3;

```

```

let industryAnalysis = '';
let careerSteps = '';
let cashSteps = '';
let dailySteps = '';

// Industry causality
if (industry === 'tech_software') {
    industryAnalysis = `With the 10-year Treasury Yield at **${macroState.us10y}%** and Fed Funds at **${macroState.fedRate}%**, the hurdle rate (discount rate) for long-duration cash flows remains restrictive. Tech companies have shifted from "growth at all costs" to "margin expansion, free cash flow (FCF), and AI operational efficiency". Speculative exploratory moonshots face cutbacks, while revenue-generating core systems, integration middleware, and AI automation infra receive concentrated capex.`;
    careerSteps = `
        * **Positioning**: Align your day-to-day deliverables directly with cost-reduction, automation, or revenue retention rather than greenfield experiments.
        * **Skill Focus**: Prioritize AI tooling integration, data pipeline resilience, and full-stack operational architecture.
        * **Job Mobility**: With unemployment at **${macroState.unemployment}%**, selective hiring exists for mission-critical senior engineers, but interviews feature rigorous proof-of-work bars. Prioritize internal promotion over risky startup moves with less than 24 months runway.
    `;
} else if (industry === 'supply_chain') {
    industryAnalysis = `Crude oil at **${macroState.oil.toFixed(2)}** and persistent tariff/nearshoring restructuring mean supply chain agility is recognized as an indispensable C-suite priority. However, higher inventory carrying costs (driven by **${macroState.fedRate}%** interest rates) penalize buffer inventory, forcing companies into multi-echelon inventory optimization (MEIO) and integrated business planning.`;
    careerSteps = `
        * **Positioning**: Lead projects optimizing safety stock calculation, supplier redundancy, and warehouse automation. Demonstrating concrete reduction in working capital unlocks rapid advancement.
        * **Skill Focus**: Master advanced planning platforms (e.g. Kinaxis, SAP IBP, O9) and dynamic pricing/tariff modeling.
        * **Job Mobility**: High defensive stability. Supply chain architects with quantitative modeling skills hold strong compensation leverage.
    `;
} else if (industry === 'finance_accounting') {
    industryAnalysis = `Yield curve dynamics (10Y at **${macroState.us10y}%**) compress M&A transaction volumes and mortgage origination, but drive record revenue for restructuring, private credit, cash treasury management, and corporate cost-audit divisions.`;
    careerSteps = `
        * **Positioning**: Pivot toward private credit analysis, liquidity management, operational

```

turnaround, and regulatory compliance.

Skill Focus: Data analytics and algorithmic forecasting to replace redundant manual audit and financial modeling work.

`;

} else {

industryAnalysis = `With CPI at  $\text{\${macroState.cpi}}$  and Fed rates at  $\text{\${macroState.fedRate}}$ , consumer spending power is divided: essential goods (healthcare, staple groceries, essential energy) maintain pricing power, while discretionary spending faces price sensitivity.`;

careerSteps = `

Positioning: Tie your responsibilities directly to bottom-line execution or customer retention.

Skill Focus: Cross-train in digital operational tools and customer lifecycle management.

`;

}

// Cash flow & Housing causality

if (housing === 'renter_saving') {

cashSteps = `

Yield Capture: With money market rates near $\text{\${(macroState.fedRate - 0.2).toFixed(2)}}$, keep your down payment fund strictly in high-yield savings (HYSAs), US Treasury Bills (4-week/13-week), or ultra-short government funds.

Mortgage Timing: The average 30-year fixed mortgage tracks the 10Y Yield plus a ~2.4% spread ($\text{\${(macroState.us10y + 2.4).toFixed(2)}}$). Do not rush into an overpriced purchase; let high rates apply downward price discovery to local seller listings while your down payment compounds risk-free.

`;

} else if (housing === 'homeowner_fixed') {

cashSteps = `

Asset Arbitrage: If your existing mortgage rate is below 4.0%, never accelerate principal payments. Instead, preserve liquidity in liquid T-bills or HYSAs earning $\text{\${(macroState.fedRate - 0.3).toFixed(2)}}$, generating positive net interest spread.

`;

} else {

cashSteps = `

Variable Debt Payoff: Variable credit cards and floating lines of credit are carrying brutal 20%+ APRs. Aggressively pay down floating debt immediately before allocating into speculative investments.

`;

}

// Daily living & purchases

```

dailySteps = `
  * **Major Purchases (Vehicles / Electronics)**: Auto loan rates are averaging 7.5% -
  10.0%. Defer non-essential car upgrades; repair and maintain existing vehicles to bypass high
  borrowing overhead.
  * **Energy & Commute**: With oil at  $\${macroState.oil.toFixed(2)}$ , baseline fuel prices are
  steady, but winter heating/cooling swings demand proactive smart-thermostat scheduling and
  grocery batching.
  * **Grocery Basket Strategy**: Food inflation has stabilized around  $\${macroState.cpi}\%$ ,
  but nominal prices remain sticky at historical peaks. Emphasize store-brand staples and bulk dry
  goods.
`;

```

```

const output = `
## 1. 🔗 Primary Causal Transmission
 $\${industryAnalysis}$ 

```

```

## 2. 📁 Career & Employability Strategy ( $\${stage.toUpperCase()}$  CAREER)
 $\${careerSteps}$ 

```

```

## 3. 💰 Cash Flow, Debt & Savings Tactic
 $\${cashSteps}$ 

```

```

## 4. 🛒 Daily Life & Living Cost Playbook
 $\${dailySteps}$ 

```

```

## 5. ⚡ 3 Concrete Actions For This Month

```

```

1. **Audit Liquid Cash Yield**: Ensure every dollar of emergency cash is in an account yielding
 $\geq \mathcal{M}acroState.fedRate - 0.4$ .

```

```

2. **Document Efficiency ROI**: Write down 2 concrete ways your day-to-day workflow saved
time, reduced cost, or stabilized revenue this quarter to present during your next performance
review.

```

```

3. **Freeze High-Cost Discretionary Debt**: Avoid financing appliances or electronics with
merchant installment loans—maintain maximum personal balance-sheet flexibility.

```

```

 $\${customQuery ? `$ 

```

```

> **Note on your inquiry ( $\${customQuery}$ )**: In the current rate regime, stability and cash
preservation give you strategic leverage. If contemplating a leap, ensure the new role is funded
by durable operational cash flow rather than short-term venture runway.` : ""}

```

```

`;

```

```

playbookOutput.innerHTML = formatMarkdown(output);
}

```

```

// 7. Everyday Dilemma Chat

```

```

sendChatBtn.addEventListener('click', handleChat);
chatInput.addEventListener('keypress', (e) => {
  if (e.key === 'Enter') handleChat();
});

async function handleChat() {
  const q = chatInput.value.trim();
  if (!q) return;

  // Add user message to UI
  appendChatMessage("You", q, "user");
  chatInput.value = "";

  // Assistant placeholder
  const aiBubble = appendChatMessage("EconPulse Assistant", "Analyzing macroeconomic
cause & effect...", "ai");

  // Prompt
  const chatPrompt = `Context: Current Fed Rate is ${macroState.fedRate}%, 10Y Yield is
${macroState.us10y}%, CPI is ${macroState.cpi}%, Unemployment is
${macroState.unemployment}%, Oil is $$${macroState.oil}.
User question: "${q}".
Provide a concise, practical, direct answer (3-4 sentences) connecting the macro cause (interest
rates/inflation/labor) to their concrete daily life or career decision.`;

  if (isModelLoaded && engine) {
    try {
      const completion = await engine.chat.completions.create({
        messages: [{ role: "user", content: chatPrompt }],
        temperature: 0.5,
        stream: false
      });
      const text = completion.choices[0]?.message?.content || "No response generated.";
      aiBubble.innerHTML = formatMarkdown(text);
    } catch (err) {
      aiBubble.innerHTML = fallbackChatAnswer(q);
    }
  } else {
    setTimeout(() => {
      aiBubble.innerHTML = fallbackChatAnswer(q);
    }, 300);
  }
}

```

```

function fallbackChatAnswer(query) {
  const q = query.toLowerCase();
  if (q.includes("car") || q.includes("vehicle") || q.includes("auto")) {
    return `With the 10-year Treasury yield at **${macroState.us10y}%**, average auto loan financing rates remain elevated around **8% to 11%**. Unless your existing vehicle requires repairs that exceed its residual value, the cause-and-effect recommendation is to maintain your current car. Financing a new purchase now locks in peak interest expense and rapid initial vehicle depreciation.`;
  }
  if (q.includes("raise") || q.includes("salary") || q.includes("negotiat")) {
    return `Headline unemployment sits at **${macroState.unemployment}%**, indicating a selective, cautious hiring environment. When negotiating compensation, avoid vague cost-of-living appeals (even with CPI at ${macroState.cpi}%). Instead, present quantifiable business metrics: how your work directly defended revenue, enhanced operational efficiency, or integrated AI productivity tools over the past 6 months.`;
  }
  if (q.includes("mortgage") || q.includes("house") || q.includes("buy")) {
    return `With the 10Y Treasury at **${macroState.us10y}%**, 30-year fixed mortgages hover near **${(macroState.us10y + 2.4).toFixed(2)}%**. While waiting indefinitely is rarely advisable if you find the right home, keep your down payment in Treasury bills earning **~${(macroState.fedRate - 0.3).toFixed(2)}%**. High rates exert downward pressure on seller asking prices, giving disciplined cash buyers growing negotiation leverage.`;
  }
  return `Given our real-time macro backdrop (Fed rate at **${macroState.fedRate}%**, CPI at **${macroState.cpi}%**), capital carries a real cost. The optimal personal strategy is to preserve liquid optionality in cash yielding over 4.5%, avoid variable revolving credit, and concentrate career efforts on high-visibility, cost-saving operational competencies.`;
}

```

```

function appendChatMessage(sender, text, type) {
  const wrapper = document.createElement('div');
  wrapper.className = "flex gap-3 text-slate-800";

  const avatar = document.createElement('div');
  if (type === 'user') {
    avatar.className = "w-7 h-7 rounded-full bg-blue-100 text-blue-700 border border-blue-200 flex items-center justify-center font-bold text-xs shrink-0";
    avatar.textContent = "U";
  } else {
    avatar.className = "w-7 h-7 rounded-full bg-emerald-100 text-emerald-800 border border-emerald-200 flex items-center justify-center font-bold text-xs shrink-0";
    avatar.textContent = "AI";
  }
}

```

```

    }

    const content = document.createElement('div');
    content.className = "flex-1 bg-white border border-slate-200 p-3.5 rounded-xl shadow-2xs";
    content.innerHTML = `
      <p class="font-bold text-slate-900 text-xs">${sender}</p>
      <div class="text-slate-700 mt-1 text-xs leading-relaxed">${text}</div>
    `;

    wrapper.appendChild(avatar);
    wrapper.appendChild(content);
    chatHistory.appendChild(wrapper);
    chatHistory.scrollTop = chatHistory.scrollHeight;
    return content.querySelector('div:last-child');
  }

  // Sample query clicks
  document.querySelectorAll('.sample-query').forEach(el => {
    el.addEventListener('click', () => {
      setActiveTab('topic-chat');
      chatInput.value = el.textContent.replace(/"/g, "");
      handleChat();
    });
  });

  // Copy & Export
  copyBtn.addEventListener('click', () => {
    const text = playbookOutput.innerText;
    navigator.clipboard.writeText(text).then(() => {
      const orig = copyBtn.textContent;
      copyBtn.textContent = "✅ Copied!";
      setTimeout(() => copyBtn.textContent = orig, 1500);
    });
  });

  exportMdBtn.addEventListener('click', () => {
    const text = playbookOutput.innerText;
    const blob = new Blob([text], { type: 'text/markdown' });
    const url = URL.createObjectURL(blob);
    const a = document.createElement('a');
    a.href = url;
    a.download = `EconPulse_Playbook_${new Date().toISOString().slice(0,10)}.md`;
  });

```

```
    a.click();
    URL.revokeObjectURL(url);
  });

refreshBtn.addEventListener('click', fetchLiveData);

// Initial Setup
checkWebGPU();
fetchLiveData();
setInterval(fetchLiveData, 30000); // 30s auto-refresh
</script>
</body>
</html>
```